Battery Threshold Calculations

1. Voltage Divider Design

Given R1 = 10kΩ and R2 = 33kΩ.

$V_{sense} = V_{bat} \times (R2 / (R1 + R2))$

$V_{sense} = V_{bat} \times (33 / 43)$

$V_{sense} = 0.767 \times V_{bat}$

2. Low Battery Threshold

Desired cutoff voltage: 3.0V

$V_{sense} = 3.0 \times 0.767 = 2.30V$

Therefore:
$V_{ref\_LOW} = 2.30V$

Comparator switches when $V_{sense} < 2.30V$.

3. Overvoltage Threshold

Desired battery limit: 4.25V

$V_{sense} = 4.25 \times 0.767 = 3.26V$

Therefore:
$V_{ref\_OV} = 3.26V$

Comparator switches when $V_{sense} > 3.26V$.

4. Battery Level Indicator Thresholds

80% Battery:
$V_{sense} = 4.0 \times 0.767 = 3.07V$

50% Battery:
$V_{sense} = 3.6 \times 0.767 = 2.76V$

20% Battery:
$V_{sense} = 3.3 \times 0.767 = 2.53V$

5. Threshold Summary

Low Battery: 3.0V → 2.30V
20% Level: 3.3V → 2.53V
50% Level: 3.6V → 2.76V
80% Level: 4.0V → 3.07V

Overvoltage: 4.25V → 3.26V

6. Hysteresis Concept

A 100kΩ positive feedback resistor is used to introduce hysteresis.

Benefits:
• Prevents output chatter near threshold voltages.
• Improves noise immunity.
• Creates separate turn-on and turn-off thresholds.
• Enables stable battery-state detection.